

HR Management & Workforce Analytics Platform

Module 7 – Operational Reporting (SSRS Design Documentation)

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1. Executive Summary

This document provides an enterprise-level design and implementation documentation for the Operational Reporting module using SQL Server Reporting Services (SSRS). The objective is to generate parameterized reports for Leave Utilization and Absenteeism tracking to support HR operational and strategic decision-making.

2. Business Context

A mid-size organization requires structured operational reporting to monitor employee leave patterns, department-level trends, and workforce availability. The SSRS reporting layer enables centralized, parameter-driven, and export-ready reports.

3. Database Architecture

The reporting module is based on the following relational schema:

3.1 Employees Table

- EmployeeId (Primary Key)
- Name
- Email
- DepartmentId (Foreign Key)
- LeaveBalance

3.2 Departments Table

- DepartmentId (Primary Key)
- DepartmentName

3.3 LeaveRequests Table

- LeaveRequestId (Primary Key)
- EmployeeId (Foreign Key)
- Status
- FromDate
- ToDate
- DaysRequested

4. Reporting View (vw_LeaveSummary)

A reporting-ready SQL View was created to aggregate leave-related data and simplify SSRS dataset mapping.

Key Fields:

- EmployeeId
- Name
- DepartmentName
- TotalLeaves
- TotalDaysTaken
- LeaveBalance

5. SSRS Reporting Architecture

Data flows from HRManagementDB to the SSRS Dataset via vw_LeaveSummary.

Parameterized filtering is applied before rendering the final report layout. Reports can be exported in PDF and Excel formats.

6. Parameterized Report Design

The following parameters are implemented:

- @StartDate
- @EndDate
- @Department
- @Employee (Optional)

Parameters enable dynamic filtering and prevent hard-coded values, ensuring enterprise scalability.

7. KPI Definitions & Calculations

7.1 Total Employees

COUNT(*) from Employees table.

7.2 Total Leave Days

SUM(DaysRequested) where Status = Approved.

7.3 Leave Utilization %

$(\text{Total Leave Taken} / \text{Total Leave Allocated}) \times 100$

7.4 Average Leave

AVG(DaysRequested) for approved leaves.

8. Report Layout Design

Top Section: KPI Summary Cards

Middle Section: Department Bar Chart & Donut Chart

Bottom Section: Monthly Leave Trend & Top 5 Employees Table

Right Panel: Filters

9. Absenteeism Calculation Logic

Absenteeism Rate = (Total Approved Leave Days / Total Working Days) × 100. This metric helps identify attendance trends and workforce gaps.

10. Data Mapping Validation

All dataset fields are accurately mapped to corresponding database tables.

Foreign key relationships ensure referential integrity.

Aggregations are validated against raw transactional data.

11. Security & Access Control

Role-based access control is implemented at the reporting layer. HR managers have access to department-wide reports, while executives access aggregated dashboards.

12. Performance Optimization

- Indexed foreign keys for optimized joins
- Pre-aggregated reporting view
- Parameter-driven filtering to reduce dataset size

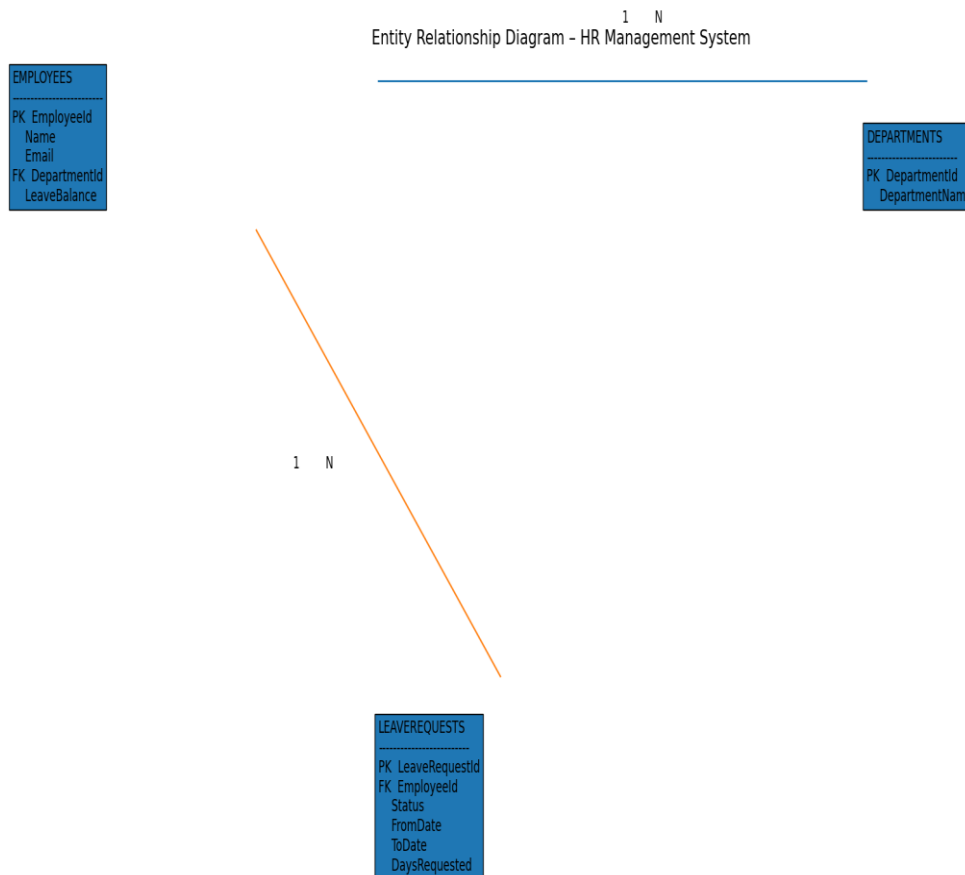
13. Enterprise Justification for SSRS

SSRS provides centralized report hosting, secure access, scheduled execution, and export capabilities, making it suitable for enterprise HR systems.

14. Future Enhancements

- Drill-through reports
- Year-over-year comparison
- Integration with Power BI
- Predictive absenteeism analytics

15: ER-Diagram



15. Conclusion

The SSRS Operational Reporting module provides structured, parameterized, and enterprise-ready HR reporting. It supports workforce tracking, leave analytics, and strategic decision-making.